

1. Image enhancement in The Spatial Domain.

- * Image enhancement improves image quality by modifying pixel values
- * Gray level transformation adjust brightness and contrast
- * Histogram processing : IT enhances image contrast using histogram equalization

* Spatial filtering Smoothing remove noise, while sharpening edges

* Computer vision use the image quality for object detection, face detection, feature extraction.

2. DIB Image Smoothing and image sharpening

Image Smoothing	Image Sharpening.
☑ Remove noise, blur	IT enhances the edge and details
IT uses mean, Gaussian and median filter	It uses Laplacian, Sobel & unsharp mask
It's best for noisy image	It's best for highlighting object boundaries IT enhance ^{edge using} the Laplacian

3. Image Enhancement Process.

- * Convert image to the frequency domain using Fourier transform
- * Low pass filter Removes high frequency noise and smooths the image.
- * High pass filter will enhance edges and the details
- * Computer vision use the low pass for image denoising and high pass for feature extraction and edge enhancement.

4. Image Segmentation

Segmentation: IT divides an image into meaningful regions or objects

Point detection IT detects isolated pixels

Line detection IT will detect straight lines.

Edge detection Object boundaries are found by edge detection

Thresholding: Separates foreground and background

Region based Similar neighboring pixels are grouped by region

Computer Vision use: IT Thresholding and edge detection are widely used for object detection and medical image analysis

5. Edge detection, edge linking and boundary detection

Edge detection IT will identify intensity changes

Edge linking IT connects broken edges to form continuous shapes

Boundary detection IT helps to extract complete object outlines